

Assignment - 07

Q1

What is meant by Object Class?

- 1) Parent class of all java classes.
- 2) Provides common methods like equals(), hashCode(), toString() & clone().
- 3) All classes in java implicitly inherit from the object class.

Q2

Explain mark & sweep algorithm of Garbage Collection!

- 1) Mark Phase: Identifies all reachable (active) objects starting from root references.
- 2) Sweep Phase: Remove unreferenced objects, freeing the memory. Helps in automatic memory management by clearing unused objects.

Q3

What are types of methods in which we can use this keyword.

- 1) Constructor - To call another constructor in same class.
- 2) Instance method - Refer to current object.
- 3) Method Parameter - To avoid naming conflicts.
- 4) Return Statement - Returns current object.

4) Explain concept of finalize() method.
 → - finalize() method, is called by garbage collector before an object is removed from memory.
 - Allows object to perform cleanup operations.
Being protected void finalized() throws Throwable {}

5) How can we call the Garbage collector explicitly in Java?
 → By using System.gc() or Runtime.getRuntime().gc().
 It requests the JVM to run the garbage collector but it is not guaranteed.

6) Diff betⁿ finalize() in Java & destructor in C++?
Finalize Destructor

Invocation	Called by Garbage Collector	Called explicitly or automatically
Usage	For cleanup before garbage collect ⁿ	Procs memory when the object goes out of scope
Syntax	protected void finalize()	~ Class name()
Multiple calls	may be called multiple times.	called only once

7) How to display the hashCode of any object in Java using hashCode() method.
 → object obj = new Object();
 System.out.println(obj.hashCode());

8) What is alternative for copy constructor in Java?
 → Cloning: Creates an exact copy of any object - use clone() method by implementing Cloneable interface.
Prog
 MyClass obj1 = new MyClass();
 MyClass obj2 = (MyClass) obj1.clone();

- Q. What is the use of import statement in java ?
- It allows access to classes from other packages.
 - Avoids writing fully qualified names.
E.g. import java.util.Scanner;

Q. What is diff betⁿ Static Block & Constructor ?

Static Block	Constructor
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Execution time	Runs once when the class is loaded.	Runs everytime an object is created.
Purpose	Initializes static variables	Initializes instance variables
Syntax	Static { }	ClassName () { }
Use Case	Per class level initialization	Per object level initialization.